Academic Statement of Purpose **University of Michigan**
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Research Interests What happens when a ML-powered machine misses a life-saving medical diagnosis? This question is central to safely deploying modern ML methods.

Specifically, classical statistical methods often provide *formal safety guarantees* in high-risk environments. However, the complex and black-box nature of modern ML methods makes providing these same assurances much harder. For instance, *diffusion models* excel at image and text generation but see limited use in critical applications, like medical image denoising, due to hallucination risks and a lack of formal guarantees.

To bridge this gap, I aim to provide **provable guarantees of reliability, privacy, and security for modern ML methods**, while maintaining strong utility, focusing on **diffusion models**. This work will allow us to deploy powerful generative models when failure is not an option.

Research Experience I began leading trustworthy ML projects as a sophomore, working with Prof. Grigorios Chrysos. My initial work focused on *unlearning* for ML classifiers, which addresses the removal of training instances from pretrained models, motivated by new legal frameworks [1]. Recent research demonstrated that even after unlearning, models often retain high-confidence predictions on most removed data [2]. I thus proposed a **test-time privacy (TTP) attack**: an adversary can use an open-weight model to confidently reinforce publicly incorrect data (e.g., a corrupted criminal record), leading to harmful decisions even after unlearning. I then introduced a **TTP defense** that protects instances within this threat model, and proved a tight privacy-utility bound. I also provided a certified algorithm without assuming convexity. Empirically, for a vision transformer trained on ImageNet, our method obtained nearly optimal protection on unsafe instances with a $< 1\%$ drop in accuracy on all others. This first-author paper was accepted in **NeurIPS'25** workshops and is currently under review at **ICLR'26** [3].

Current Focus After making progress on improving the safety of classifiers, I have since expanded my work to **understanding and mitigating diffusion hallucinations**. As observed by Aithal et al. [4], most diffusion hallucinations are interpolations between the target modes. Recently, I proved a theorem explaining this phenomenon for a Gaussian mixture. I am now working to leverage these insights to mitigate hallucinations.

Research Vision Safety verification is crucial for user trust, but few provably correct algorithms exist. I believe this is due to inconsistent mathematical definitions across trustworthy ML. For example, formalizing hallucination for generative image models is challenging; one current definition is generating low-probability instances. However, target distributions are often heavy-tailed, which means low-probability instances can be valid.

Furthermore, these definitions often differ across modalities. For example, memorization in text generation is often defined at the *concept* level (reproducing information like "Paris is the capital of France"), whereas in image generation it is often defined at the *instance* level (e.g., based on pixel similarity).

Thus, a key goal of my Ph.D. is to **provide clear and unified mathematical definitions across modalities for unlearning, hallucination, and memorization**. For example, to unify the definitions of memorization between text and image modalities, one could leverage causal inference to define memorization as a training instance having an excessively strong causal link to the generated data. I will then create *verification algorithms* for these formal definitions, supplying regulators with tools to audit legal compliance, reliability, and user privacy.

Role of UMich At UMich, I am excited to work with **Prof. Qing Qu**, extending his lab's

recent work on the probability flow distance (PFD), which can be used to formally study diffusion generalization and memorization errors [5]. Specifically, I propose leveraging the PFD to quantify diffusion generalization error under distribution shift ($\mathcal{E}_{\text{shift}}$). I then aim to study how $\mathcal{E}_{\text{shift}}$ changes under different denoising architectures (e.g., UNets vs. transformers) to derive insights on diffusion model selection under distribution shift.

Next, I am eager to work with **Prof. Samet Oymak**. His lab's recent work connects learning a self-attention layer to estimating the transition matrix of a Markov chain, which is then leveraged to provide sample complexity bounds and characterize empirical phenomena like text repetition [6]. I propose extending this to diffusion language models (dLMs) with transformer denoisers; learning a transformer denoiser will be equivalent to learning the transition matrix of a similar Markov chain, which instead evolves across diffusion steps. I then propose studying this Markov chain's dynamics to derive the optimal dLM noising schedule, constrained to ensure that the attention co-occurrence graph remains connected throughout the reverse process. Since generation quality critically depends on the noise schedule, I believe this will significantly reduce dLM hallucination rates.

Finally, I am interested in working with **Prof. Changxiao Cai**, extending his lab's recent work on characterizing the optimal convergence rate of dLMs as $O(\frac{1}{T})$, where T is the number of iterations in the reverse process [7]. I propose providing a similar minimax rate for dLMs under differential privacy to better understand the privacy-utility tradeoff in the diffusion setting.

My long-term goal is to become a tenure-track professor advancing trustworthy ML. The mentorship available at UMich, with Profs. Qu, Oymak, and Cai, positions me ideally to achieve this.

References

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